Software-Based Firewall Integration with Wazuh SIEM

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1. Task Summary

Objective:

Implement a firewall rule on Kali Linux using IPSet to block all incoming traffic from Sweden-based IP addresses, monitor this activity using Wazuh SIEM, and validate the effectiveness of the block with alert logs.

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2. Tools and Setup

Lab Environment

Component	Description
OS	Kali Linux (VirtualBox)
Firewall Tool	IPSet + IPTables
Monitoring Tool	Wazuh Agent + Wazuh Manager
Target IP Source	IPDeny Sweden IP List
Network Mode	Bridged Adapter

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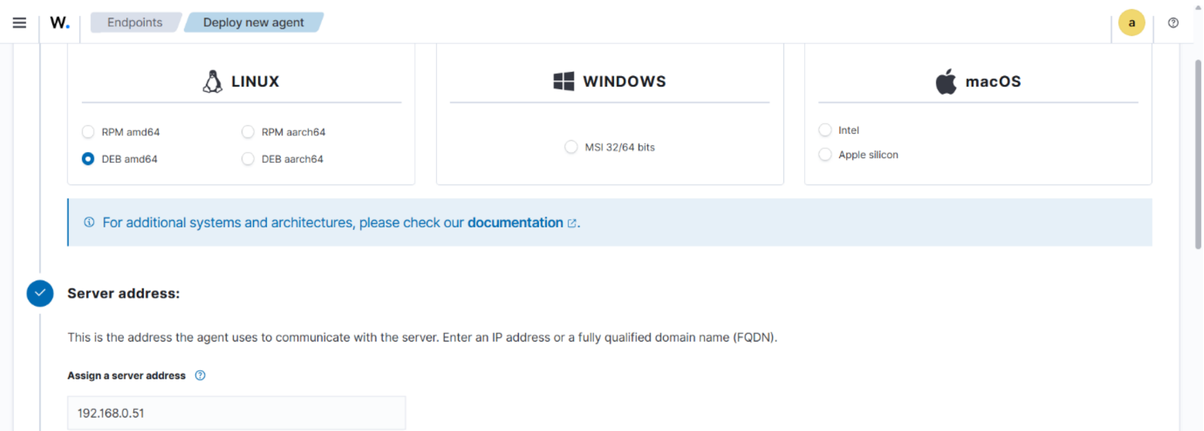
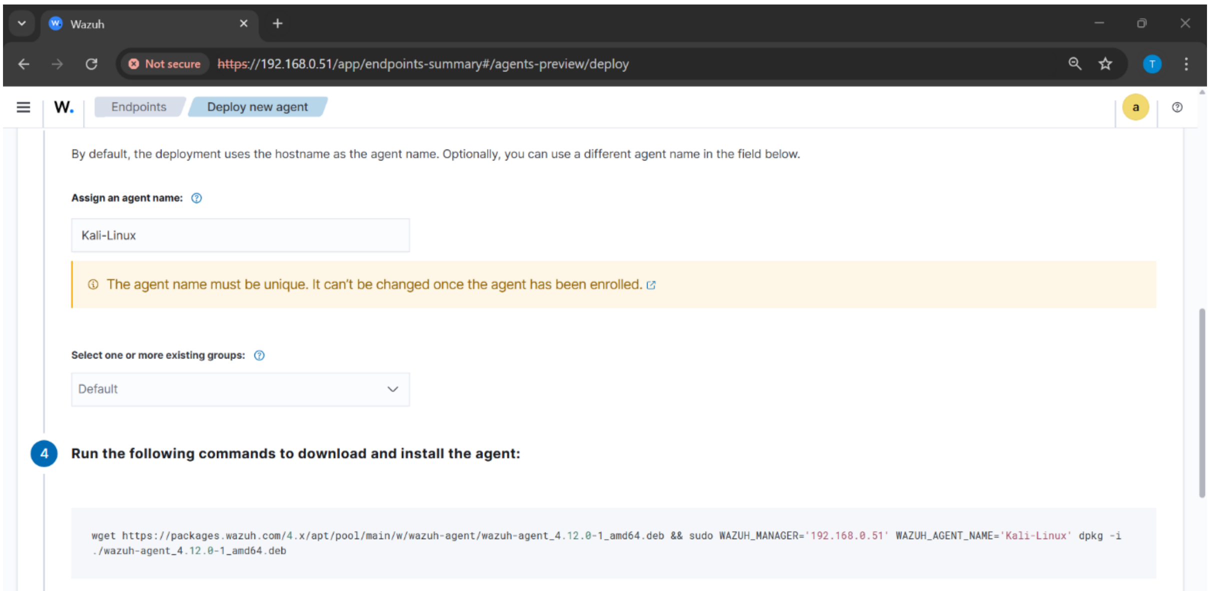
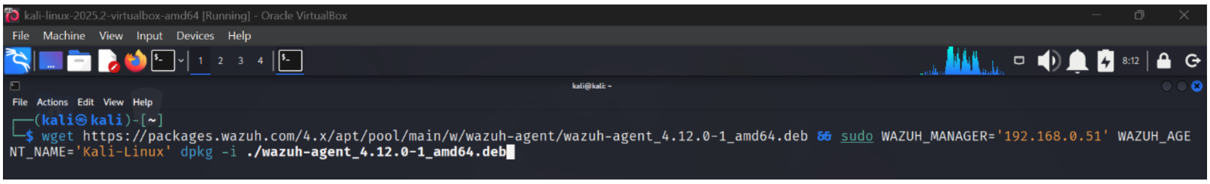
3. Step-by-Step Implementation:

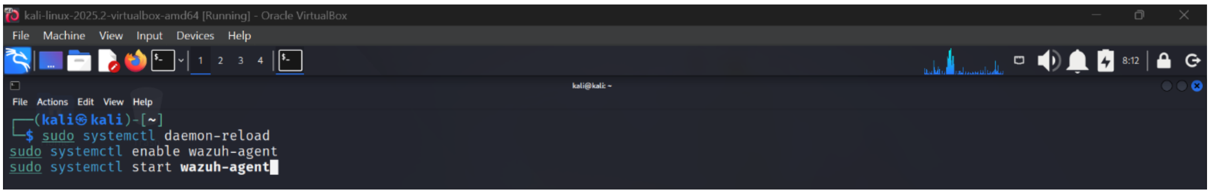
A. Deploy Wazuh-agent into Kali Linux



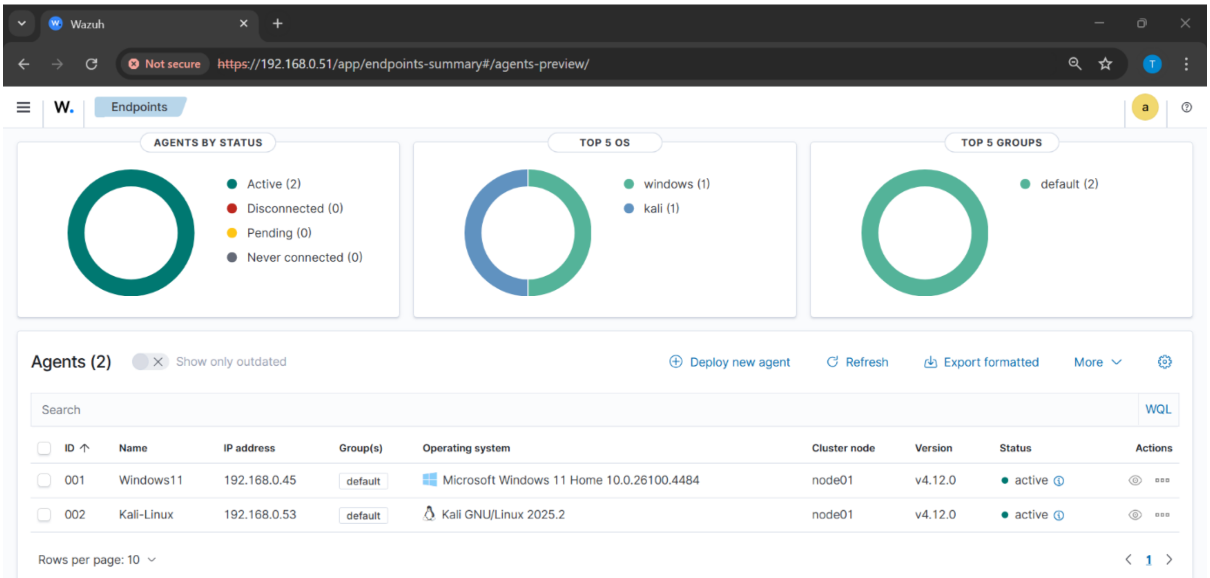


B. Copy Paste the Commands into Kali Linux Terminal:



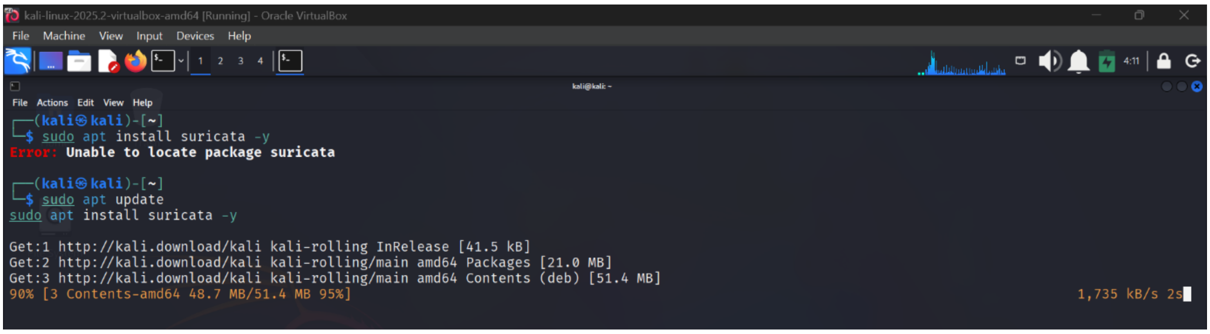


Agent has been connected properly:

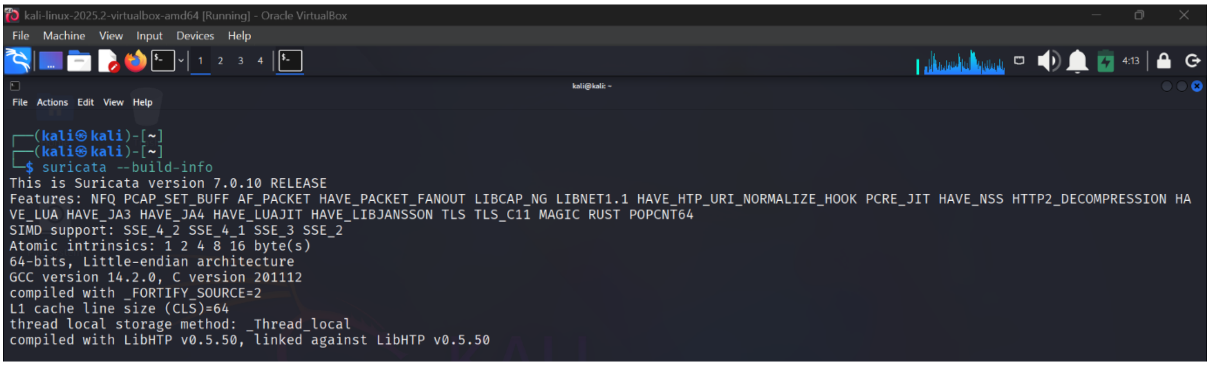


Step 2: Now Install Suricata into Kali Linux

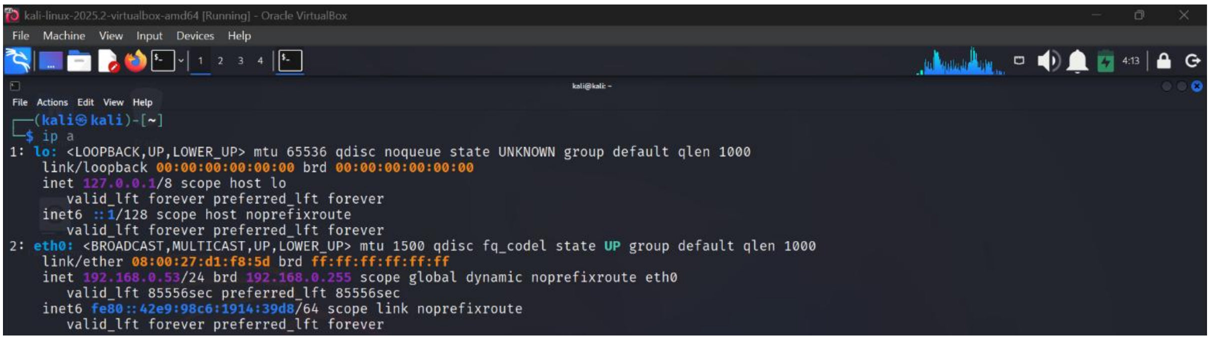
A. Update Kali & Install Suricata:



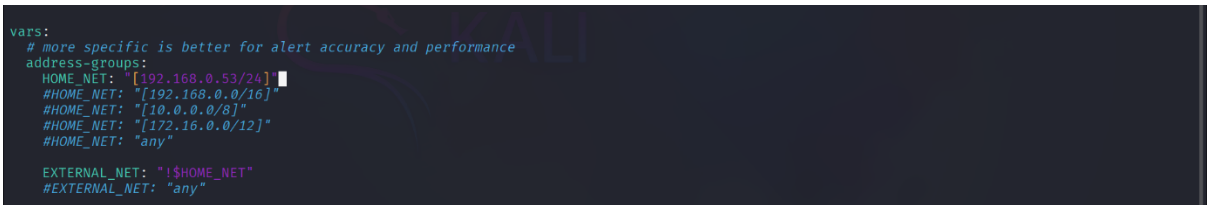
B. Check the info & Version:



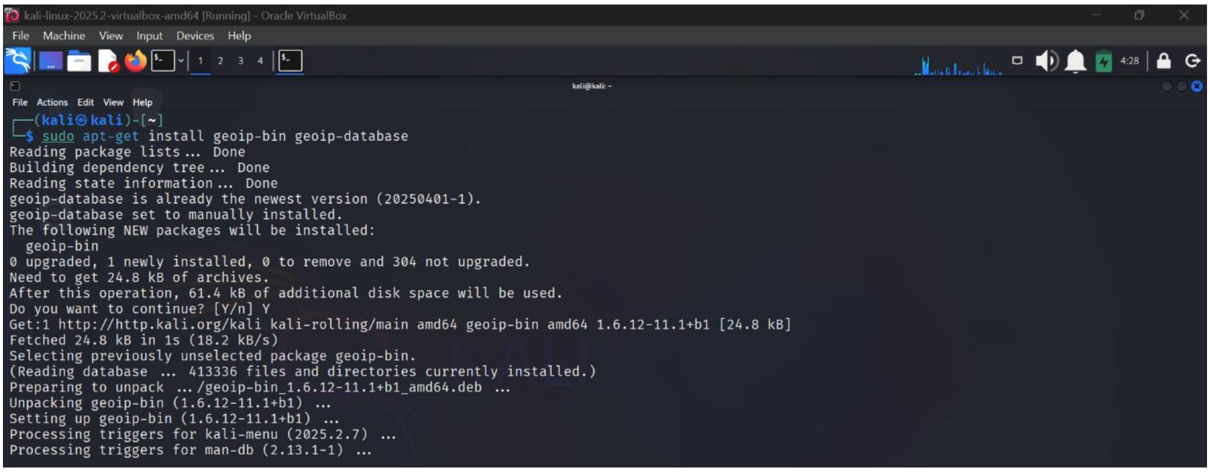
C. Check your Kali OS IP:



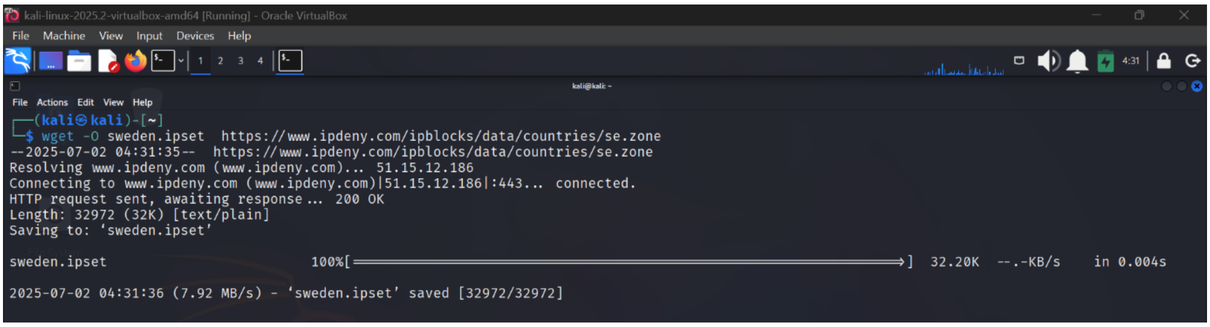
D. Add you ip addr:



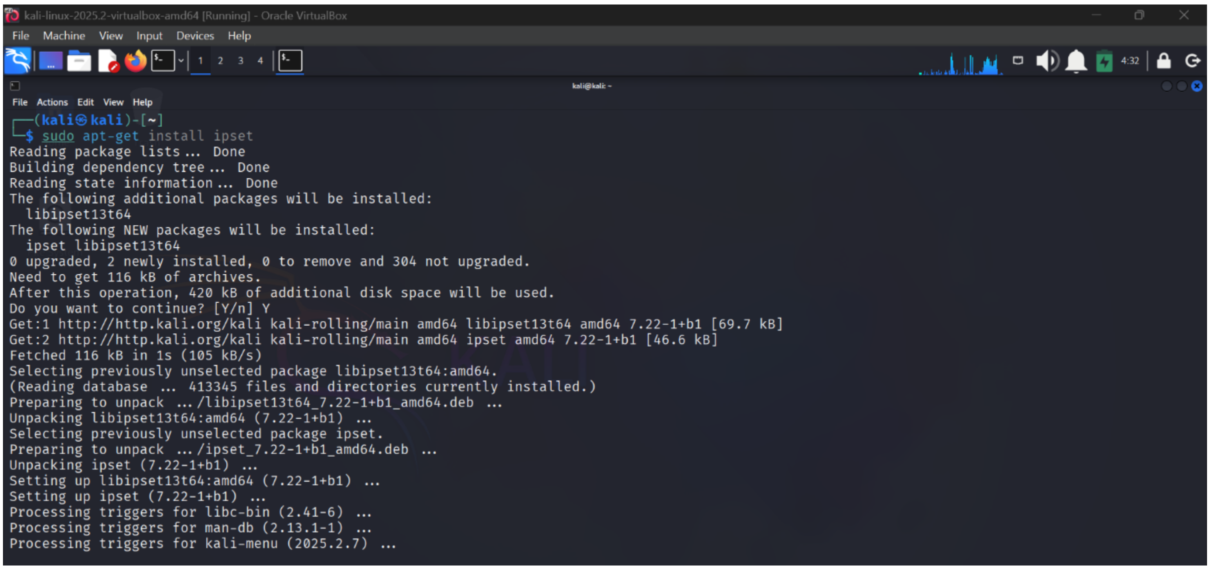
E. Installing GeoIP Tools for Country-Based Traffic Filtering on Kali Linux:



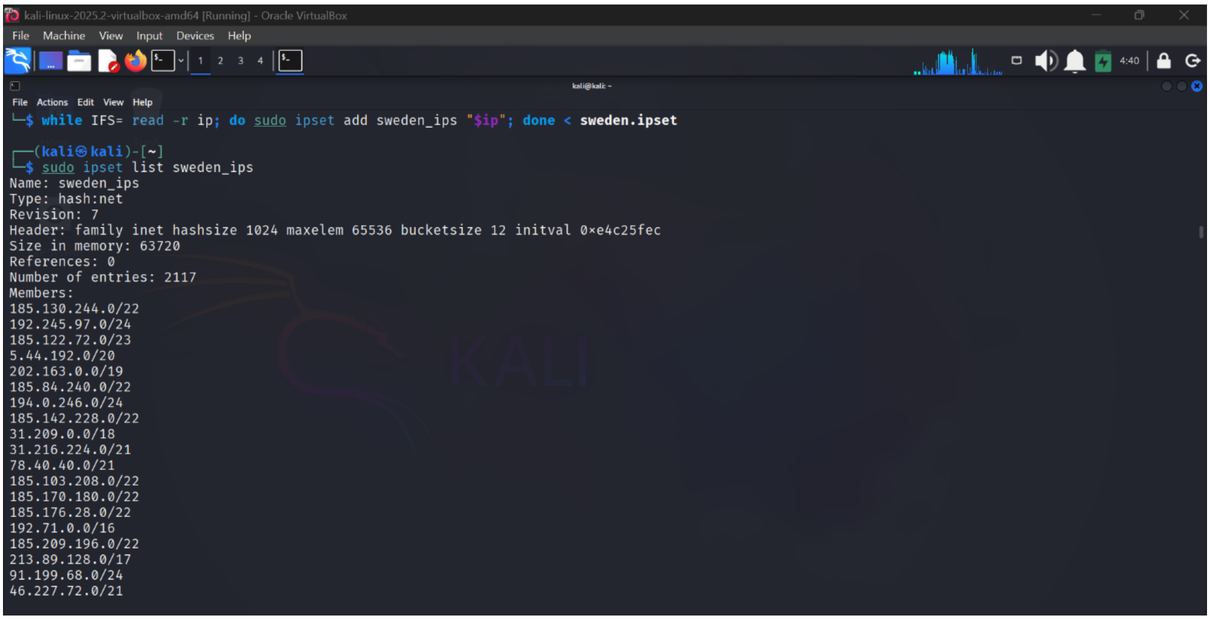
F. Now Download Sweden.ip set:



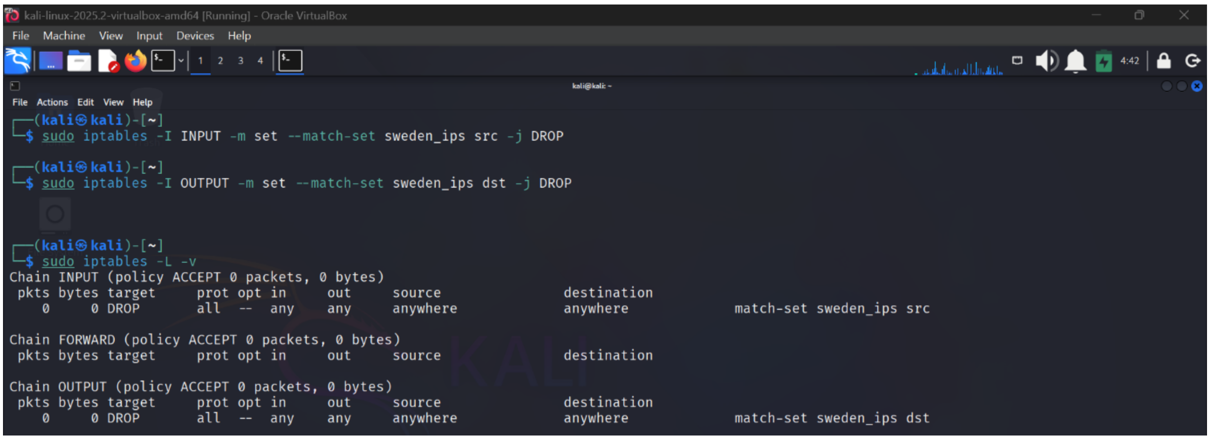
G: Setting Up IPSet for Geo-Based Blocking:



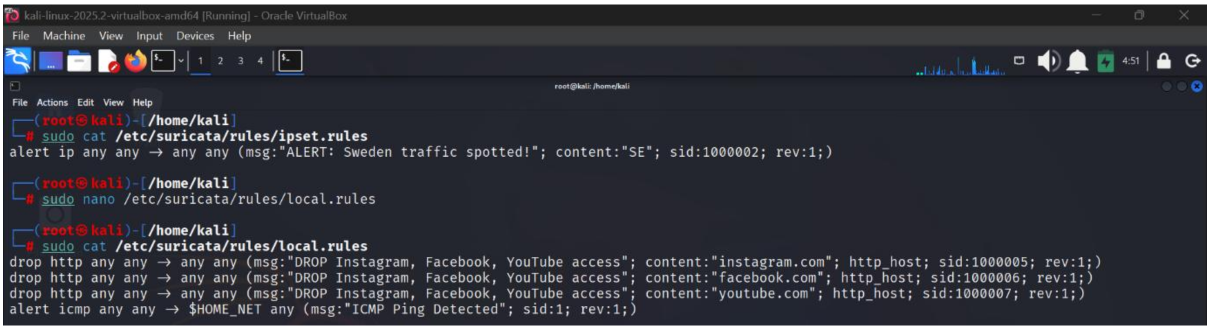
Step 3: Add IPs to IPSet from File Verify IPSet



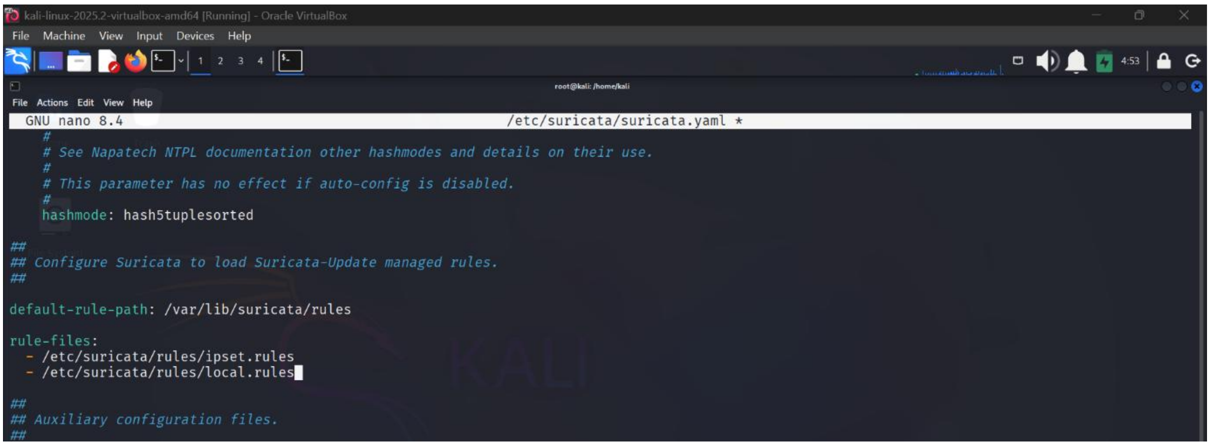
Step 5: Block Traffic from Sweden



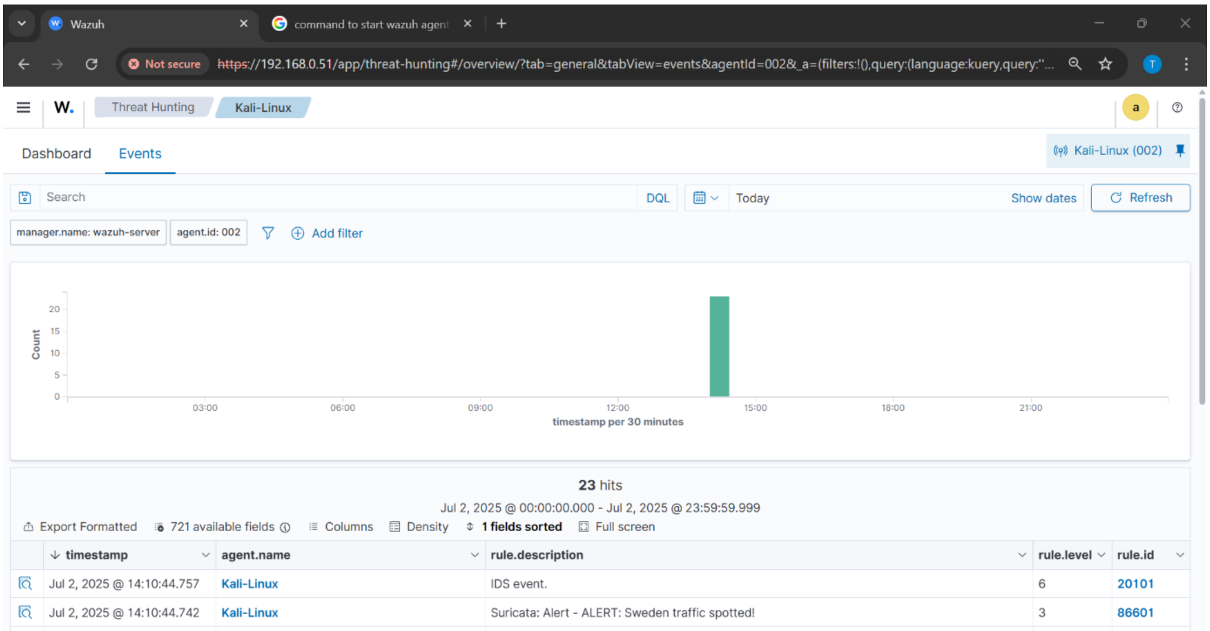
Step 6: Adding Rules

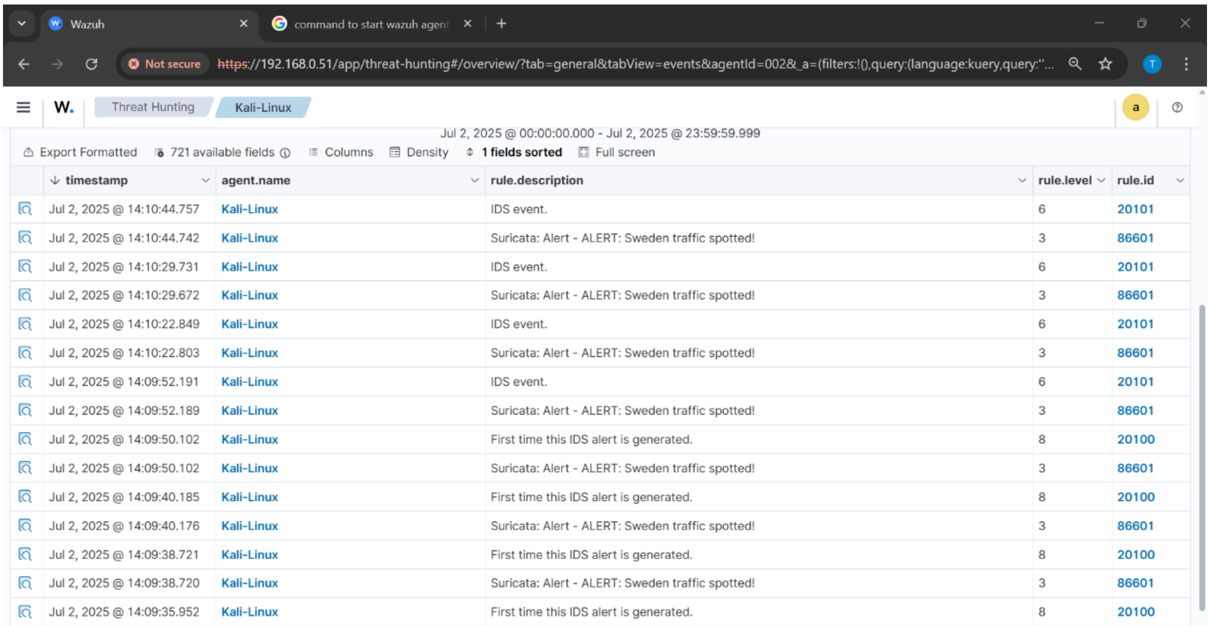


Step 7: Configuring OSSEC to monitor Suricata logs

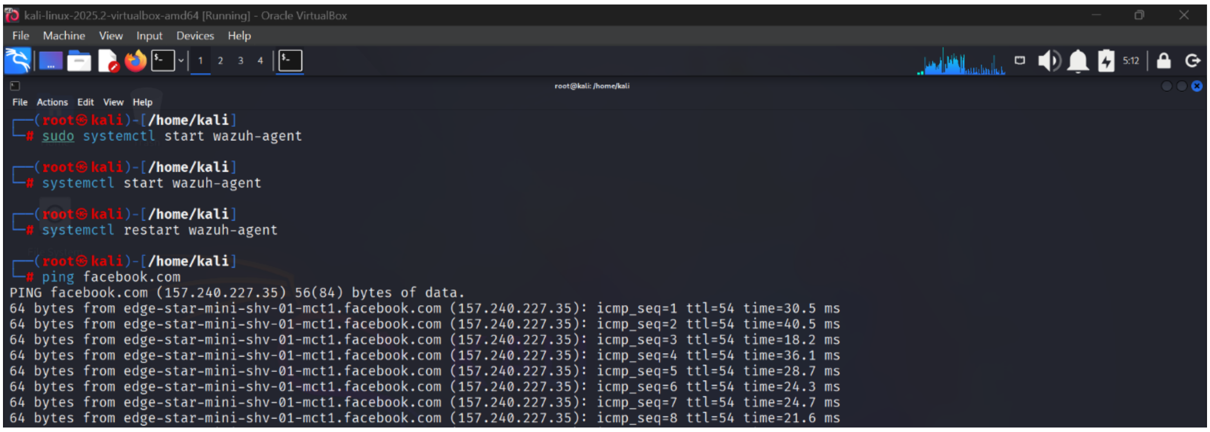


Step 8: Monitor in Wazuh Dashboard

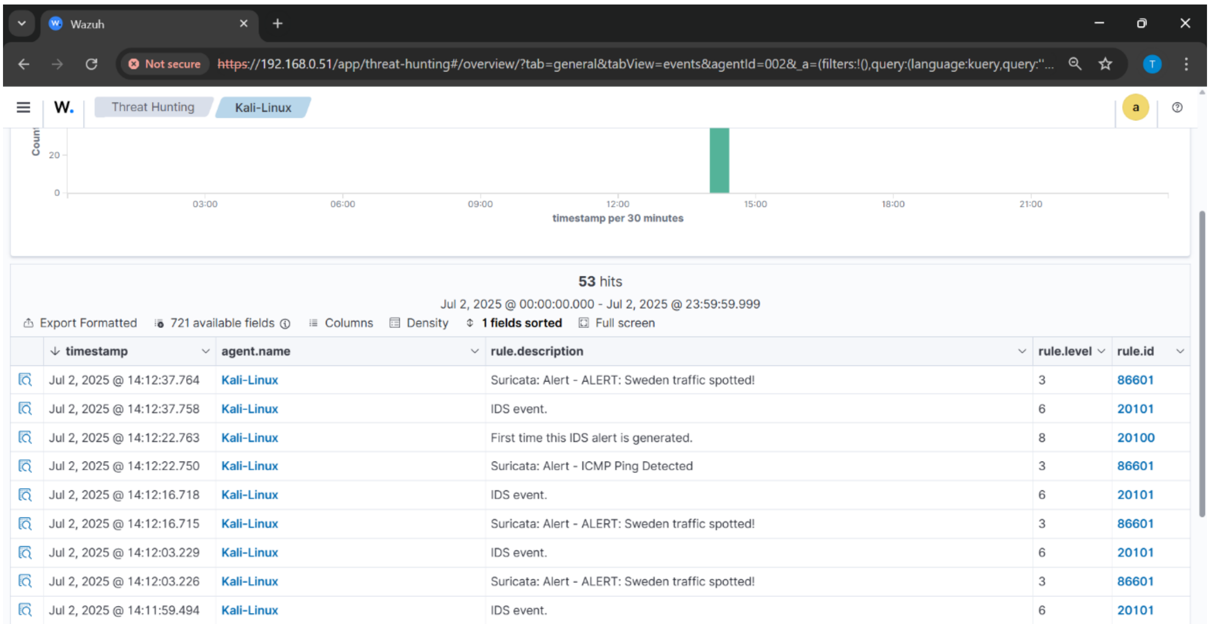




Using Ping command to verify the logs:



Here are logs after ping to Facebook:



Validation Tests:

Test Scenario	Result
Ping/traceroute from Sweden IP	Blocked/Dropped
Alert visibility in Wazuh	Successful
IPSet integrity verified	Successful

Observations & Findings:

- Sweden IPs successfully blocked at firewall level using IPSet.
- Suricata/Wazuh Agent can be extended to monitor those IPs.
- Wazuh dashboard reflected system-level alerts (if logs monitored).
- Highly efficient method for mass-country IP blocking with low resource usage.

Conclusion:

Successfully implemented IPSet-based blocking for Sweden IP ranges and validated the rule through traffic blocking and alert monitoring in Wazuh. This method provides lightweight country-based control with full SOC visibility.